THE RELATIVE MOR FUNCTOR INTO ANY TARGET IS AN ALGEBRAIC SPACE FOR FINITE LOCALLY FREE SOURCE

THE STACKS PROJECT AUTHORS

Editorial note. Footnote 'Some details omitted' covers the set-theoretic size verification that the coproduct of the F_E is an algebraic space; retained verbatim and flagged for the reviewer.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 1709 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Base change of Mor. lemma-base-change-hom-functor (same chapter, named). Makes the etale reduction on B possible.
- Etale cover of Mor. lemma-etale-covering-hom-functor (same chapter, named). Makes the etale reduction on X possible.
- Mor into affine. more-morphisms-lemma-hom-from-finite-free-into-affine (cross-chapter, named). Provides algebraicity of each F_E.
- Glueing spaces. spaces-lemma-glueing-algebraic-spaces (cross-chapter, named). Assembles the open covering into an algebraic space.

Source reference. Original reference 04S6: focused-seeds/geometry/S9-crit-hom-functor-algebraic-space/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05Y4: sources/stacks-criteria/pinned-original-source.tex, source lines 1590–1590. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05Y5: sources/stacks-criteria/pinned-original-source.tex, source lines 1612–1612. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05Y6: focused-seeds/geometry/S9-crit-hom-functor-algebraic-space/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the

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Source reference. Original reference 02WL: focused-seeds/geometry/S9-crit-hom-functor-algebraic-space/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02WR: focused-seeds/geometry/S9-crit-hom-functor-algebraic-space/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Proposition 0.1. *Let S be a scheme. Let $Z \to B$ and $X \to B$ be morphisms of algebraic spaces over S. If $Z \to B$ is finite locally free then $Mor_B(Z, X)$ is an algebraic space.*

Proof. Choose a scheme $B' = \coprod B'_i$ which is a disjoint union of affine schemes B'_i and an étale surjective morphism $B' \to B$. We may also assume that $B'_i \times_B Z$ is the spectrum of a ring which is finite free as a $\Gamma(B'_i, \mathcal{O}_{B'_i})$-module. By Lemma 05Y4 and Spaces, Lemma 02WL the morphism $Mor_{B'}(Z', X') \to Mor_B(Z, X)$ is surjective étale. Hence by Bootstrap, Theorem 04S6 it suffices to prove the proposition when $B = B'$ is a disjoint union of affine schemes B'_i so that each $B'_i \times_B Z$ is finite free over B'_i. Then it actually suffices to prove the result for the restriction to each B'_i. Thus we may assume that B is affine and that $\Gamma(Z, \mathcal{O}_Z)$ is a finite free $\Gamma(B, \mathcal{O}_B)$-module.

Choose a scheme X' which is a disjoint union of affine schemes and a surjective étale morphism $X' \to X$. By Lemma 05Y5 the morphism $Mor_B(Z, X') \to Mor_B(Z, X)$ is representable by algebraic spaces, étale, and surjective. Hence by Bootstrap, Theorem 04S6 it suffices to prove the proposition when X is a disjoint union of affine schemes. This reduces us to the case discussed in the next paragraph.

Assume $X = \coprod_{i \in I} X_i$ is a disjoint union of affine schemes, B is affine, and that $\Gamma(Z, \mathcal{O}_Z)$ is a finite free $\Gamma(B, \mathcal{O}_B)$-module. For any finite subset $E \subset I$ set

$$F_E = Mor_B(Z, \coprod\nolimits_{i \in E} X_i).$$

By More on Morphisms, Lemma 05Y6 we see that F_E is an algebraic space. Consider the morphism

$$\coprod\nolimits_{E \subset I \text{ finite}} F_E \longrightarrow Mor_B(Z, X)$$

Each of the morphisms $F_E \to Mor_B(Z, X)$ is an open immersion, because it is simply the locus parametrizing pairs (a, b) where b maps into the open subscheme $\coprod_{i \in E} X_i$ of X. Moreover, if T is quasi-compact, then for any pair (a, b) the image of b is contained in $\coprod_{i \in E} X_i$ for some $E \subset I$ finite. Hence the displayed arrow is in fact an open covering and we win[1] by Spaces, Lemma 02WR. □

[1]Modulo some set theoretic arguments. Namely, we have to show that $\coprod F_E$ is an algebraic space. This follows because $|I| \leq \text{size}(X)$ and $\text{size}(F_E) \leq \text{size}(X)$ as follows from the explicit description of F_E in the proof of More on Morphisms, Lemma 05Y6. Some details omitted.